Skewness & Kurtosis — SciPy Summary

Quick revision notes for Data Science: meaning, practical use, SciPy code, interpretation, and key points.

1. Skewness

Skewness measures the asymmetry of a distribution. It tells us whether the data has a longer tail on the left or right side.

Value	Meaning
> 0	Positive / Right Skewed — longer tail on the right
< 0	Negative / Left Skewed — longer tail on the left
≈ 0	Approximately symmetric

2. Kurtosis

Kurtosis describes the tail behavior of a distribution and helps indicate the presence of extreme observations. In SciPy, the default is Fisher's definition.

Fisher Kurtosis	Interpretation
> 0	Leptokurtic — relatively heavier tails / more extreme observations
< 0	Platykurtic — relatively lighter tails
≈ 0	Mesokurtic — similar to the normal distribution

3. SciPy Code

```
import numpy as np
from scipy.stats import skew, kurtosis

data = np.array([10, 12, 15, 18, 20, 22, 25, 30, 35, 80])

s = skew(data)
k = kurtosis(data)

print("Skewness:", s)
print("Kurtosis:", k)
```

4. Fisher vs Pearson Kurtosis

SciPy's **kurtosis(data)** uses Fisher's definition by default. A normal distribution has Fisher kurtosis ≈ 0. With **fisher=False**, SciPy returns Pearson kurtosis, where a normal distribution has kurtosis ≈ 3.

```
print(kurtosis(data)) # Fisher: normal ≈ 0
print(kurtosis(data, fisher=False)) # Pearson: normal ≈ 3
```

5. Bias Parameter

bias=True is the default. Use **bias=False** when you want the bias-corrected sample statistic.

```
skew(data, bias=False)
kurtosis(data, bias=False)
```

6. Where Are They Used?

• Exploratory Data Analysis (EDA) — understand distribution shape.

• Data cleaning — investigate unusual or extreme observations.

• Outlier analysis — identify distributions with long/heavy tails.

• Finance — returns, income, transaction amounts.

• Banking — transaction behavior and unusual values.

• Healthcare — measurements and patient-related numerical data.

• Machine Learning — decide whether a transformation or further distribution analysis may be useful.

7. Practical Data Science Workflow

Load Data → Clean Data → EDA → Mean/Median → Variance/Std → Skewness/Kurtosis → Investigate Outliers/Distribution → Transform if needed → Machine Learning

8. Most Important Code to Remember

```
from scipy.stats import skew, kurtosis

skew(data)
kurtosis(data)

kurtosis(data, fisher=False)

skew(data, bias=False)
kurtosis(data, bias=False)
```

■ Easy Memory Trick

Skewness = Which SIDE is the tail?
Kurtosis = How heavy/extreme are the tails?

Use these measures mainly to understand a distribution during EDA; do not calculate them blindly for every column.